

Thiranex Data Analytics Internship Task 4

Data Cleaning & Reporting Automation

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1. Introduction

This project focuses on automating the cleaning and preprocessing of a raw coffee sales dataset to convert inconsistent transactional data into a structured, analysis-ready format suitable for reporting and visualization.

2. Objective

The main aim was to improve data quality by handling missing values, removing inconsistencies, and standardizing the dataset to support efficient analysis and automated reporting.

3. Methodology

The dataset was cleaned using Python by addressing missing values in numeric columns with median imputation and categorical columns with mode values. Duplicate records were removed, and inconsistencies in formats such as dates and categories were standardized. Data types were also corrected to ensure consistency and accuracy across all fields.

4. Outcome

The final output is a clean and structured dataset with improved accuracy and consistency. The preprocessing workflow reduces manual effort and enables faster generation of reports and visual insights, making the dataset suitable for further analysis and dashboard creation.

5. Conclusion

This project demonstrates how automated data cleaning improves efficiency and reliability in analytics workflows. By standardizing and preparing raw data, it becomes easier to generate meaningful insights and support data-driven decision-making.

